



COEP TECHNOLOGICAL UNIVERSITY

(COEP TECH)

(Formerly College of Engineering Pune)

T1 Examination

(CT-20014) Data Structures and Algorithms – II

Course: B.Tech, Semester IV

Branch: Computer Engineering

Academic Year: 2023-2024

Max.Marks:20

Duration: 1 Hours

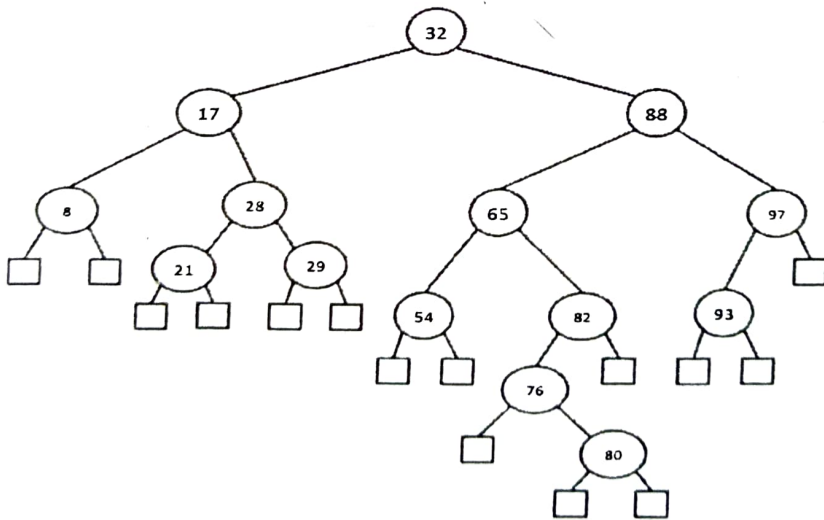
Date: 17/02/2024

Instructions:

Student MIS No.

612203021

- Figures to the right indicate the full marks.
- Mobile phones and programmable calculators are strictly prohibited.
- Writing anything on question paper is not allowed.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Write your MIS Number on Question Paper

			Marks
Q 1	a	Given an integer n, explain how many numbers of structurally unique binary search trees will be there which has exactly n nodes of unique values from 1 to n.	2
	b	If number of nodes 'n' are given in binary tree then minimum and maximum height of the tree will be _____ and _____ respectively.	2
Q 2	a	Consider the binary search tree below. Draw the binary search tree obtained after performing remove (88). Justify your answer for each solution. 	2
	b	Let the preorder traversal sequence of binary tree T be 100, 34, 16, 9, 8, 38, 11, 4, 81 and postorder traversal sequence be 34, 9, 11, 4, 38, 81, 8, 16, 100. If all the non-leaf nodes of T have two children, identify T. Mention all steps.	3

Q3	a Write the output of the following function: <pre> void function1(tree t) { node *p; queue q; qinit(&q); enq(&q, t); while(!qempty(&q)) { p=deq(&q); printf("%d",p->data); if(p->left) enq(&q, p->left); if(p->right) enq(&q, p->right); } printf("\n"); } </pre> The tree t is a binary search tree, having data inserted in the given order: 75, 68, 89, 21, 72, 80, 96. Make suitable assumptions wherever required.	2
	b Explain construction of Expression tree to evaluate given postfix expression - Input: s = ["4","2","+","3","5","1","-","*","+"]	3
Q4	a Write a function to display the path from root to the farthest leaf of a binary tree of integers. Assume linked implementation of the tree. Also assume all required data structures are available.	3
	b Write an iterative function in C to insert node in a BST using linked list. Given a root node reference of a BST and a key, insert the node with the given key in the BST.	3